An FPGA Ring-Oscillator-Based True Random Number Generator with von Neumann Conditioning and AXI-Stream Output

Rahul Nerella, Uttiyo Majumder, and Kevin Velanjickal
Department of Electronics and Communication Engineering
Vellore Institute of Technology, Vellore, India
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***Abstract*—This paper documents a parameterized FPGA true random number generator (TRNG) implemented in SystemVerilog. The design combines a bank of ring-oscillator (RO) entropy sources, clock-domain sampling, a pair-wise von Neumann corrector, a word packer, and a first-word-fall-through FIFO with AXI-Stream-style valid/ready signaling. The archived RTL defaults to four five-stage oscillators, 16-bit output words, and a 16-word FIFO. Functional verification was performed in ModelSim using a behavioral Bernoulli/jitter source rather than zero-delay combinational ROs. The latest archived self-checking simulation reports five passing phases: a fixed-pattern scoreboard, bias removal, nibble-uniformity sanity check, backpressure/integrity, and reset recovery. For a separate 8,528-bit simulated stream, the archived NIST SP 800-22 summary reports 10 passes, one failure, and four ineligible tests. The reported Random Excursion status is inconsistent with the scalar p-value displayed in the same summary, and therefore requires revalidation. No physical FPGA capture, synthesis resource report, or hardware entropy characterization is included; accordingly, these results establish simulation-level functional behavior only, not silicon-level entropy quality or cryptographic suitability.**

***Index Terms*—true random number generator, FPGA, ring oscillator, entropy source, von Neumann corrector, clock-domain crossing, AXI-Stream, NIST SP 800-22.**

I. Introduction

Random-bit generation is a basic building block in hardware security, cryptographic provisioning, simulation, and randomized testing. A deterministic pseudorandom generator expands an initial state, while a physical TRNG attempts to derive unpredictable bits from a physical process. In an FPGA, timing variation and jitter in free-running ring oscillators can be sampled against a system clock to form an entropy source. The quality of that source depends on the actual device, implementation, sampling relationship, environmental conditions, and conditioning path; a simulation model cannot prove that physical entropy is present.

The Experiment 1 project develops a complete RTL data path from raw entropy to a word-oriented interface. The stages are (i) an RO bank, (ii) synchronization into a system-clock domain, (iii) von Neumann post-processing, (iv) bit-to-word packing, and (v) FIFO buffering with backpressure. A self-checking testbench covers protocol and algorithmic behaviors, and an archived bitstream dump is analyzed with a subset of the NIST SP 800-22 statistical test suite [1].

The main contributions of this project are a modular SystemVerilog implementation, integration of a simple bias-removal algorithm, an output queue with a valid/ready handshake, and repeatable simulation checks. The limits are equally important: the statistical testbench uses a behavioral random-bit model, only 8,528 bits were used in the archived NIST run, and no physical capture or FPGA implementation results are supplied.

II. System Architecture

A. End-to-End Data Path

Fig. 1 is a corrected architecture drawing derived from the source modules. The hardware path uses the RO bank; the testbench selects a behavioral source through the `USE_RING` parameter. Both paths enter the same sampled-bit, correction, packing, and buffering stages.

B. Ring-Oscillator Entropy Source

The module `entropy_source.sv` has parameters `N_OS`=4 and `OSC_STAGES`=5 by default. Each generated loop uses a gated first stage and a chain of inverter stages, and the oscillator outputs are combined into `osc_raw`. Attributes such as `keep` are used to discourage synthesis optimization of the feedback loop. Ring oscillators are intentionally combinational loops and require tool-specific constraints and implementation review; the source constraints are marked as templates and their pin assignments must be checked against the actual board's master constraints.

In the simulation configuration, the testbench sets `USE_RING=0` and drives `ent_override_i`. This avoids attempting to simulate a zero-delay combinational loop as a physical oscillator. The generated test stream is therefore evidence about the downstream digital conditioning and interface, not evidence of on-chip oscillator jitter.

C. Sampling and Conditioning

The asynchronous raw source is sampled into the system-clock path by `cdc_sync.sv`. The design flags CDC hardening as an area for review. In the provided RTL, the two-bit internal `ff_meta` register is followed by `ff[0]` when `STAGES=2`; the effective serial pipeline depth should be

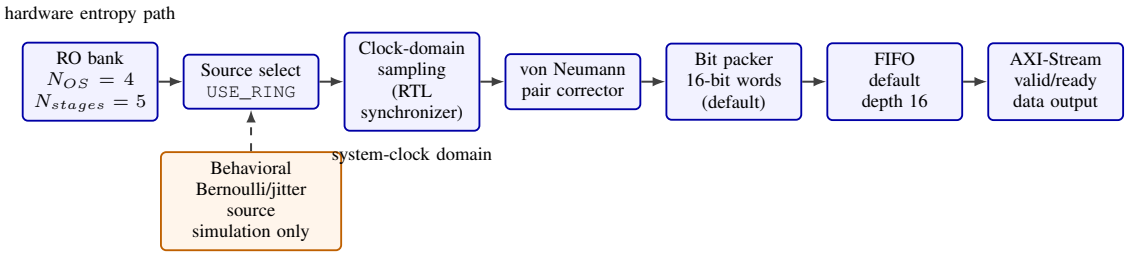


Fig. 1. TRNG architecture reconstructed from `trng_top.sv` and its submodules. The dashed source is a simulation substitute, not a physical entropy source.

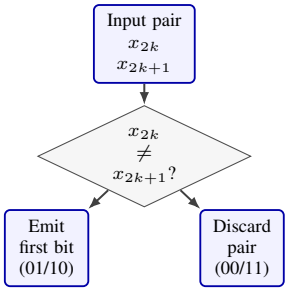


Fig. 2. Pairwise von Neumann correction used by the RTL. Pairing is non-overlapping.

reconciled with the intended CDC depth and latency before hardware sign-off.

The von Neumann corrector consumes non-overlapping pairs. For independent identically distributed input bits with $P(1) = p$, an accepted pair occurs only when the pair is 01 or 10. Since

$$P(01) = P(10) = p(1-p), \tag{1}$$

the probability of an output one, conditioned on a pair being accepted, is

$$P(Y = 1 \mid \text{accepted}) = \frac{P(10)}{P(01) + P(10)} = \frac{1}{2}. \tag{2}$$

This guarantee relies on independence and stationarity assumptions; it does not remove arbitrary temporal correlation, and it does not establish that an FPGA source provides enough entropy. For a pairwise implementation, `01` emits one value, `10` emits the other value, and `00`/`11` emit no output, as shown in Fig. 2.

D. Word Packer and FIFO Interface

The packer shifts each accepted corrected bit into a register and pulses `wren` after a complete word. The default top-level configuration uses 16-bit words, with the first collected bit positioned at the most significant side of the completed word. The FIFO exports `tvalid`, `tready`, and `tdata`; its `tdata` path is combinational (first-word-fall-through style). Writes arriving while full are dropped, and a one-cycle `overflow` indication is exposed. The testbench treats overflow as a failure. For a real consumer, handshake timing and overflow behavior must be integrated deliberately rather than assumed to be lossless under arbitrary backpressure.

TABLE I
PRINCIPAL RTL MODULES

Module	Responsibility
`entropy_source`	Four gated five-stage RO loops; OR-combines outputs.
`cdc_sync`	Samples asynchronous raw bit into the system-clock path.
`vn_corrector`	Non-overlapping pairwise correction; suppresses equal pairs.
`trng_top`	Selects source, packs bits into words, connects the FIFO.
`trng_fifo`	Small queue with valid/ready signaling and overflow pulse.

TABLE II
DEFAULT RTL PARAMETERS VS. FUNCTIONAL TEST CONFIGURATION

Parameter	RTL default	Testbench
RO count	4	bypassed
RO stages	5	bypassed
Output word width	16 bits	8 bits
FIFO depth	16 words	8 words
Entropy source	RO bank	behavioral input
System clock	100 MHz intent	100 MHz model

III. RTL ORGANIZATION AND DESIGN PARAMETERS

Table I summarizes the principal source files. The default design parameters should not be confused with the smaller testbench configuration.

The design files also contain board constraint templates for Basys 3 and Nexys A7. The package does not establish that either board was selected for final implementation. The board-specific XDC files explicitly require pin-level verification before synthesis and must not be treated as fully validated constraints.

IV. VERIFICATION METHODOLOGY

A. Functional Testbench

The self-checking testbench `tb_trng_top.sv` drives a 100 MHz simulated system clock and replaces the RO source with a controllable pseudo-random/Bernoulli bit generator. The five phases are summarized in Fig. 3.

The testbench phases are:

1) *Fixed-pattern scoreboard:* a repeating raw-bit pattern is sent through the DUT and compared against two

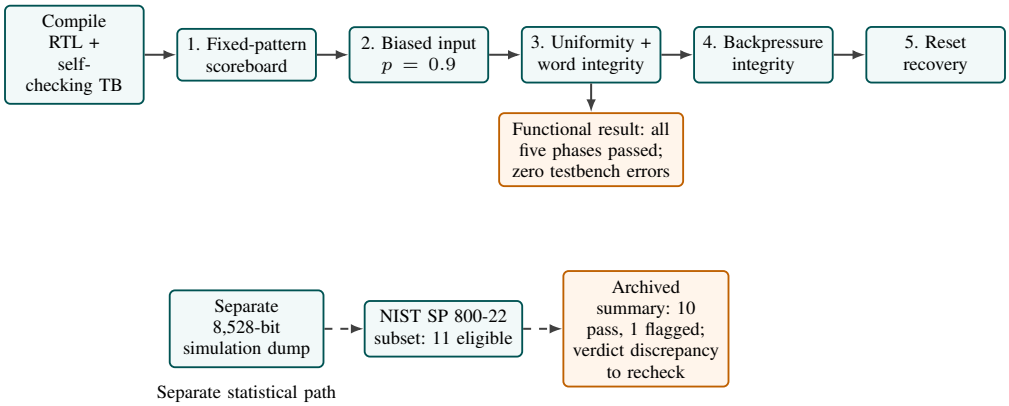


Fig. 3. Verification strategy. Functional checks and the separate NIST analysis are different evidence tracks and are not merged into a single claim.

possible reference-pair alignments; a match of at least 95% is required.

2) *Bias removal:* a source with $p = 0.9$ is applied; the emitted-one fraction must remain between 0.40 and 0.60.
3) *Integrity and uniformity:* after a reset, deterministic data is checked for consistent output words and the uniform-source phase computes a nibble histogram chi-square statistic with a testbench sanity limit of 45.
4) *Backpressure:* the sink temporarily stalls and drained words are compared with a mirror scoreboard to detect loss, duplication, corruption, or reordering.
5) *Reset recovery:* reset is asserted during operation; the FIFO is expected to become invalid and the pipeline must resume producing output.

The included fresh-rerun log states that compilation had zero errors and two pre-existing SystemVerilog-check warnings in `entropy_source.sv`; the simulation ended with `ALL CHECKS PASSED`. The warnings concern ModelSim's checking support around `always_comb`; the archived log does not demonstrate a failed functional assertion.

B. NIST SP 800-22 Analysis

A separate archived stream contains 8,528 output bits, with 4,239 ones and an overall ones ratio of 0.4971. The accompanying `nist_summary.txt` lists 11 eligible tests, four ineligible tests, ten PASS labels, and one FAIL label. The result is too small to characterize the physical RO source and should be considered exploratory, particularly because the bits were generated by a software model.

The archived summary labels Random Excursion as failed with scalar $p = 0.038930$. The supplied p-value chart marks significance at 0.01, which would not classify that scalar p-value as below the threshold. The source artifacts therefore contain an unresolved mismatch between the suite's status label and the plotted scalar value. The original status is preserved in Table IV, but the Random Excursion verdict should not be treated as resolved until the suite is rerun and its state-level/proportion output is inspected.

TABLE III
FUNCTIONAL SIMULATION RESULTS FROM THE ARCHIVED TRANSCRIPT

Check	Observed result
Fixed-pattern reference	DUT produced 509 corrected outputs; best reference match was 509/510 (phase gate passed).
Bias removal ($p = 0.9$)	1,727 valid corrected bits; 854 ones; fraction 0.4945 (inside 0.40–0.60 bound).
Word integrity	187 words written and 187 drained in the deterministic integrity phase; no scoreboard errors reported.
Nibble uniformity ($p = 0.5$)	1,066 words / 2,132 nibbles; reported $\chi^2 = 9.055$ with a sanity limit of 45.
Backpressure	7 words packed and 7 drained; content comparison passed.
Reset recovery	FIFO invalid after reset and output recovered within the test timeout.
Overall	`ALL CHECKS PASSED`; zero testbench errors in the archived run.

V. SIMULATION RESULTS

A. Functional Results

Table III records the values in the ModelSim transcript. The functional results support control/data-path behavior in the simulated source configuration; they do not establish physical unpredictability.

Fig. 4 shows the saved 16-bin nibble histogram. The reported testbench statistic is 9.055; the historical image labels the statistic as approximately 9.1 with 15 degrees of freedom and a 95% critical value of 25.0. This is a sanity check for gross defects in the simulated packing/conditioning chain, not a cryptographic randomness certification.

B. Bit Balance and NIST Results

The running ones-ratio plot in Fig. 5 starts with a transient and approaches approximately 0.5 over the displayed 8,528-bit stream. The aggregate source summary gives an exact ratio of 0.4971. Balance alone is not sufficient to establish randomness: correlations, predictability, and other structural patterns must also be tested.

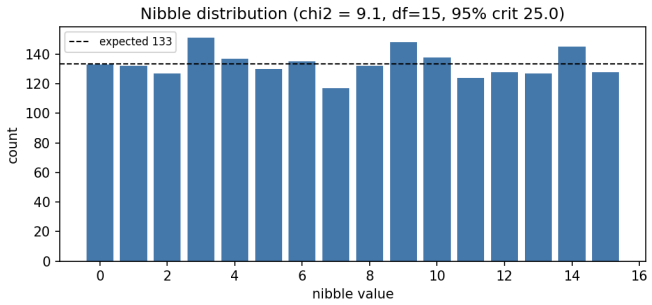


Fig. 4. Archived output-nibble histogram from the behavioral simulation stream.

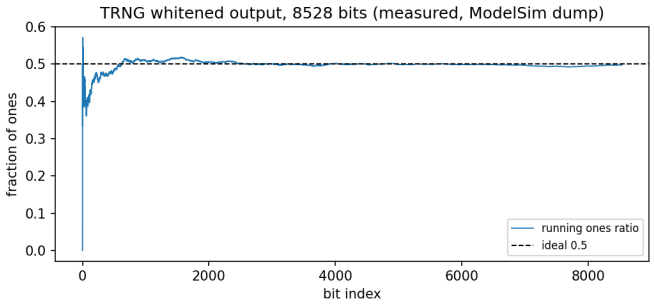


Fig. 5. Running fraction of ones over the 8,528-bit simulated dump.

Fig. 6 reproduces the p-value plot supplied in the project archive. Table IV gives the explicit archived status labels. The original plot and text summary should be read with the mismatch discussed above; the report does not infer a corrected verdict without a fresh suite run.

VI. Hardware Implementation Considerations

The repository includes XDC templates for Basys 3 and Nexys A7, but does not identify a final board selection or include a validated implementation report. Before hardware deployment, the following actions are necessary:

- Select the exact board and FPGA part, then verify every pin against the official board master XDC.
- Constrain clocks and review asynchronous paths; ensure the RO loops are preserved by the synthesis flow and excluded from ordinary synchronous timing analysis where appropriate.
- Inspect the synthesized netlist to confirm that every intended oscillator remains, and record the actual cell/resource and timing reports.
- Review the synchronizer implementation, placement attributes, and reset behavior for the selected device.
- Add runtime health checks and an explicit failure/recovery policy. The current FIFO exposes dropped-word overflow but does not guarantee losslessness if the consumer stalls until the queue is full.
- Capture a sufficiently large bitstream directly from the FPGA, retain acquisition metadata, and run statistical tests on physical output. A behavioral testbench dump must never be presented as hardware output.

The report does not provide a resource-utilization table because no genuine synthesis result was supplied. Any rough estimate based on RTL size would not be a substitute for the FPGA tool report. The actual pin mappings also remain unverified because the constraint files in the repository are marked as templates.

VII. Applications and Limitations

Potential uses include random challenge generation, randomized test inputs, simulation seeds, and entropy input for a larger security subsystem, but cryptographic use would require a validated entropy source, a documented conditioning strategy, startup and continuous health tests, and a security review. The implementation as currently evidenced should be treated as an educational FPGA TRNG prototype.

The main limitations are:

1) **No hardware entropy measurement:** testbench bits are generated by a software-controlled model, not captured from the ring oscillators.
2) **Small statistical sample:** the archived NIST sequence is only 8,528 bits, and four tests were ineligible.
3) **Unresolved NIST status mismatch:** Random Excursion's text status and reported scalar p-value require a fresh run and detailed inspection.
4) **CDC implementation review:** the effective serial depth in the current RTL should be checked against intended synchronizer staging.
5) **FIFO overflow policy:** writes are dropped when full, so a sustained stall can cause data loss.
6) **No synthesis/timing evidence:** the repository does not contain verified area, Fmax, power, or bitstream results.

VIII. Conclusion

A modular SystemVerilog TRNG datapath was assembled from a ring-oscillator source, sampled-bit pipeline, von Neumann corrector, packer, and FIFO interface. The archived ModelSim transcript reports successful fixed-pattern, bias-removal, nibble-distribution, backpressure, and reset-recovery checks. Under the behavioral input model, the biased-input test emitted ones at a fraction of 0.4945, and the nibble sanity statistic was 9.055. The separate 8,528-bit statistical run is preliminary: its summary records ten NIST passes and one flagged failure, and the Random Excursion verdict is inconsistent with the displayed scalar p-value. No physical FPGA capture, synthesis report, or hardware entropy characterization is available. Consequently, the project demonstrates an RTL-level TRNG pipeline and a useful functional verification harness, but the remaining statistical inconsistency and hardware validation must be resolved before any claim of true physical randomness or security suitability.

Acknowledgment

This work is documented as Experiment 1 of the V-SPACE FPGA Build Challenge 2026. Project source, testbenches,

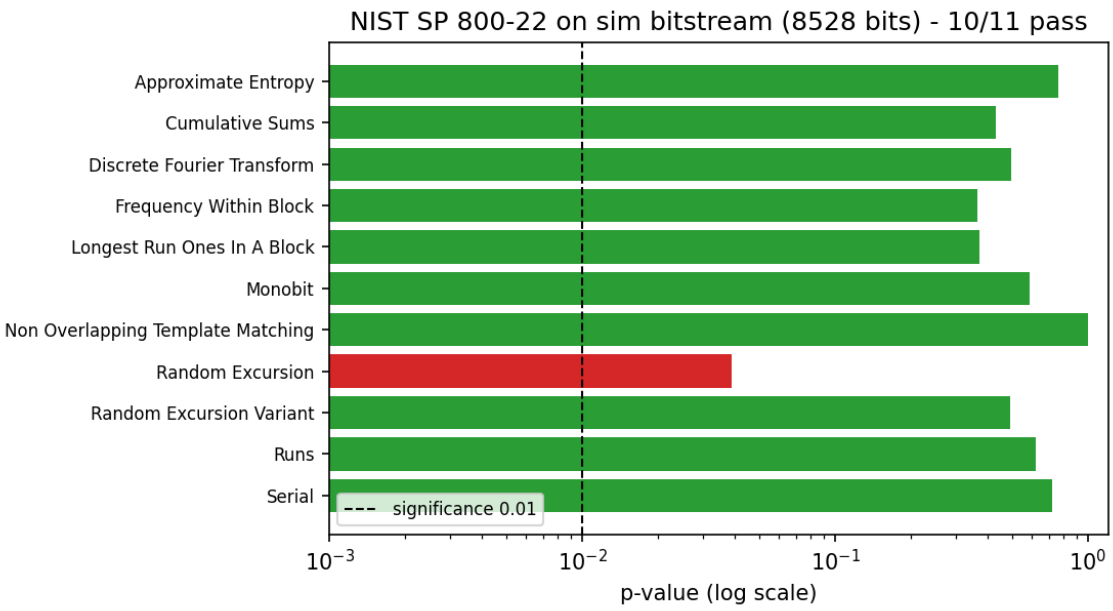


Fig. 6. Archived p-values for the 11 eligible tests; dashed line is the significance level drawn in the source plot.

TABLE IV
NIST RESULTS AS RECORDED IN SIMULATION/NIST_SUMMARY.TXT

Test	Status	p-value	Test	Status	p-value
Approximate Entropy	PASS	0.761417	Random Excursion	FAIL*	0.038930
Cumulative Sums	PASS	0.431771	Random Excursion Variant	PASS	0.492089
Discrete Fourier Transform	PASS	0.499217	Runs	PASS	0.620626
Frequency Within Block	PASS	0.366085	Serial	PASS	0.722988
Longest Run Ones in a Block	PASS	0.372729			
Monobit	PASS	0.588208			
Non-overlapping Template Matching	PASS	0.999466			

*The archived text summary labels Random Excursion FAIL despite showing $p = 0.038930$, which is above the 0.01 significance level in the archived plot. Re-run the original test suite and inspect full output before resolving this discrepancy. Overall archived count: 10 PASS, 1 FAIL, 0 errors among 11 eligible tests; 4 other tests were ineligible.

transcripts, and plots were taken from the supplied TeamFPGA repository archive. The figures and results are identified as archived simulation artifacts where applicable.

REFERENCES

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